

Q13 $n \gg 0$ ($n+1$ tosses for Mary, n tosses for John) $|\Omega| = 2^{2n+1}$
 $\Omega = \{H, T\}^{2n+1}$

Prob. Mary has ~~at least~~ more heads than John. $= \{H, T\}^{n+1} \times \{H, T\}^n$

[Aim: Prob = $\frac{1}{2}$] [Sub-aim: # ways for Mary to get more head than John = 2^{2n}]

Induction

~~Base case: $n=0$ is trivial~~

~~Inductive case:~~

Exp. $n=3 \Rightarrow \Omega = \{H, T\}^4 \times \{H, T\}^3$

$((H, H, T, T), (T, T, H)) \in \Omega$

!!

$(M, J) \in \Omega$, M is an $(n+1)$ -tuple
 J is an n -tuple

we simply write

HHTT for a seq.

↑ obtained by Mary.

Induction:

$n=0$ is trivial

Suppose the result holds true for $n=k$, for some $k \geq 0$.

ways ... $= 2^{2k}$.

Let $(P, Q) \in \cancel{\{H, T\}^k} \times \{H, T\}^{k+1}$.

- $P = (p_1, p_2, \dots, p_k)$ is an k -tuple representing a sequence of head and tails tossed by John.

- Q is an $(k+1)$ -tuple ... by Mary.

Now add 1 coin at the end of P and Q .

$P[] \quad Q[]$

Consider $P' = (P, x)$, where $x \in \{H, T\}$, $Q' = (Q, y)$ where $y \in \{H, T\}$.

Case $|P| < |Q|$
John Mary

Let p be the # head in P .

Let q be the # head in Q .

Case $\overset{k \text{ coins}}{p} < \overset{k+1 \text{ coins}}{q} \Rightarrow$ There are 2^{2k} possibilities for (P, Q) .
(by inductive hypothesis)

• $(x, y) = (H, H) \Rightarrow$ # head in P'
 (T, T) $<$ # head in Q' .

possibilities (P', Q') such that $(x, y) = (H, H)$ and

head in $P' < \dots Q' = \boxed{2 \cdot 2^{2k}} \quad (T, T)$

• $(x, y) = (T, H) \Rightarrow$ Trivial ~~$\Rightarrow$ It contributes~~

$\Rightarrow$ It contributes $\boxed{2^{2k}}$ possibilities.

$$(x, y) = (H, T)$$

We want to count # ways where John has more ~~head~~ or equal head to Mary.

$$(\underbrace{p+1}_{\text{John}} \geq \underbrace{q}_{\text{Mary}})$$

Claim. $p+1 = q$

If $p+1 > q \Rightarrow p > q$ (Contradict to $p < q$)

Total possibilities

$$\underbrace{\hspace{1cm}}_{2^k}$$

= # ways that Mary has more head

This is what we want

+ # ways that John has more or equal head

~~ways~~ possibilities of (p, q) s.t. $p+1 = q$.

[p varies from 0 to k]

poss. s.t. $p+1=q$

= # poss. ^{of (p, q)} that John has $\overset{p}{\textcircled{0}}$ heads, Mary has $\overset{q}{\textcircled{1}}$ head

+ # poss. ^{(p, q)} that John has $\textcircled{1} \dots \dots \dots \textcircled{2}$ head

$\vdots$

+ # poss. $\dots \dots \dots \textcircled{k} \dots \dots \dots \textcircled{k+1}$ head

$$= \sum_{i=0}^k \# \text{ poss. } \dots \underbrace{i \text{ heads}}_{\binom{k}{i}} \dots \underbrace{i+1 \text{ heads}}_{\binom{k+1}{i+1}}$$

$$= \sum_{i=0}^k \binom{k}{i} \binom{k+1}{i+1}$$

$\therefore$ # ways that Mary ... more head

$$= 2^{2k} - \sum_{i=0}^k \binom{k}{i} \binom{k+1}{i+1}$$

Case $p \geq q$

~~$p < q$~~ For $p > q$, no ~~any~~ possibilities ~~would have~~ where Mary has more heads.

p varies from 0 to k
For $p = q$, when $(x, y) = (H, H), (T, T), (H, T)$,
no possibilities.

When $(x, y) = (T, H)$, every possibilities ~~are valid~~.
have more heads ~~to~~ for Mary.

possibilities of (p, q) where $p = q$

$$\begin{aligned} &= \sum_{i=0}^k \underbrace{\binom{k}{i}}_{\text{John}} \underbrace{\binom{k+1}{i}}_{\text{Mary}} = \sum_{i=0}^k \binom{k}{k-i} \binom{k+1}{k+1-i} \\ &= \sum_{j=0}^k \binom{k}{j} \binom{k+1}{j+1} \end{aligned}$$

$$\underline{\binom{a}{b} = \binom{a}{a-b}}$$
$$j = k - i$$

$$\# \text{ Total} = 2 \cdot 2^{2k} + 2^{2k} + \left(2^{2k} - \sum_{i=0}^k \binom{k}{i} \binom{k+1}{i+1} \right) + \sum_{j=0}^k \binom{k}{j} \binom{k+1}{j+1}$$

$$= 4 \cdot 2^{2k} = 2^{2k+2} = 2^{2(k+1)}$$

$\therefore$ By induction, $\# \dots = 2^{2n}$ for $n \geq 0$.

$$\therefore \text{Prob.} = \frac{\# \dots}{|\Omega|} = \frac{2^{2n}}{2^{2n+1}} = \frac{1}{2}.$$

2nd Approach

John has n coins,

Mary has $n+1$ coins.

$$\begin{aligned} \# \text{ ways to get } k \text{ heads from } n \text{ coins (for John)} \\ = \binom{n}{k} \end{aligned}$$

$$\begin{aligned} \# \text{ ways to get } m \text{ heads from } n+1 \text{ coins (for Mary)} \\ = \binom{n+1}{m} \end{aligned}$$

$$\begin{aligned} \# \text{ ways where John gets } k \text{ heads and Mary gets} \\ m \text{ heads} = \binom{n}{k} \binom{n+1}{m} \end{aligned}$$

$$\begin{aligned} \# \text{ ways where Mary has more heads than John} \\ = \sum_{k=0}^n \sum_{\boxed{m > k}} \binom{n}{k} \binom{n+1}{m} = \sum_{k=0}^n \sum_{m=k+1}^{n+1} \binom{n}{k} \binom{n+1}{m} \end{aligned}$$

$$\therefore \# \text{ ways} = \sum_{j=0}^n \left[\binom{n}{j} \sum_{i=j+1}^{n+1} \binom{n+1}{i} \right] =: S_n$$

• Check base case

Claim. $S_n = 4^n$ for $n \geq 1$. [Use induction: Suppose $S_k = 4^k$.]

$$S_{k+1} = \sum_{j=1}^{k+1} \binom{k+1}{j} \sum_{i=j+1}^{k+2} \binom{k+2}{i}$$

$$\sum_{j=0}^k \binom{k}{j} \sum_{i=j+1}^{k+1} \binom{k+1}{i}$$

$$\begin{aligned} \sum_{i=j+1}^{k+2} \binom{k+2}{i} &\stackrel{\text{Pascal}}{=} \sum \binom{k+1}{i} + \sum \binom{k+1}{i-1} \\ &= \left[2 \sum_{j=j+1}^{k+1} \binom{k+1}{i} \right] + \binom{k+1}{j} \end{aligned}$$

$$\sum \binom{k+1}{i} = \binom{k+1}{j+1} + \binom{k+1}{j+2} + \dots + \binom{k+1}{k+1} + \boxed{\binom{k+1}{k+2}} \quad \text{Zero}$$

$$\sum \binom{k+1}{i-1} = \binom{k+1}{j} + \binom{k+1}{j+1} + \dots + \binom{k+1}{k} + \binom{k+1}{k+1}$$

$$\therefore S_{k+1} = \sum_{j=0}^{k+1} \boxed{\binom{k+1}{j}} \left[2 \sum_{i=j+1}^{k+1} \binom{k+1}{i} + \binom{k+1}{j} \right]$$

~~↖~~ ↓ Pascal

$$= \sum_{j=0}^{k+1} \left[\binom{k}{j} + \binom{k}{j-1} \right] \left[2 \sum_{i=j+1}^{k+1} \binom{k+1}{i} + \binom{k+1}{j} \right]$$

$$= \underbrace{\sum_{j=0}^{k+1} \binom{k}{j} [\dots]}_{\text{1st Sum}} + \underbrace{\sum_{j=0}^{k+1} \binom{k}{j-1} [\dots]}_{\text{2nd Sum}}$$

1st Sum:

$$\sum_{j=0}^{k+1} \binom{k}{j} \left[2 \sum_{i=j+1}^{k+1} \binom{k+1}{i} + \binom{k+1}{j} \right]$$

$$= 2 \underbrace{\sum_{j=0}^{k+1} \binom{k}{j} \sum_{i=j+1}^{k+1} \binom{k+1}{i}}_{S_k = 4^k} + \sum_{j=0}^{k+1} \binom{k}{j} \binom{k+1}{j} = \boxed{2 \cdot 4^k + \sum \binom{k}{j} \binom{k+1}{j}}$$

2nd sum:

$$\sum_{j=0}^{k+1} \binom{k}{j-1} [2 \dots]$$

$$= \cancel{2} \sum_{\boxed{j=0}}^{k+1} \binom{k}{j-1} \sum_{i=j+1}^{k+1} \binom{k+1}{i} + \sum_{\boxed{j=0}}^{k+1} \binom{k}{j-1} \binom{k+1}{j}$$

$$j=0, \binom{k}{j-1} = \binom{k}{-1} \stackrel{\text{def}}{=} 0$$

$$= 2 \sum_{\boxed{j=1}}^{k+1} \binom{k}{\boxed{j-1}} \sum_{i=\boxed{j+1}}^{k+1} \binom{k+1}{i} + \sum_{j=1}^{k+1} \binom{k}{j-1} \binom{k+1}{j}$$

Replace ~~j-1~~ by j'

$$j=1 \Rightarrow j'=0$$

$$j=k+1 \Rightarrow j'=k$$

$$\# \quad 2 \sum_{j'=0}^k \binom{k}{j'} \sum_{i=j'+2}^{k+1} \binom{k+1}{i} = 2 \sum_{j=0}^k \binom{k}{j} \sum_{i=j+2}^{k+1} \binom{k+1}{i}$$

$$= 2 \sum_{j=0}^k \binom{k}{j} \sum_{i=j+1}^{k+1} \binom{k+1}{i} - 2 \sum_{j=0}^k \binom{k}{j} \binom{k+1}{j+1} = \sum_{i=j+1}^{k+1} \binom{k+1}{i} - \binom{k+1}{j+1}$$

$$S_k = 4^k$$

$$2nd \text{ sum} = 2 \cdot 4^k - 2 \sum_{j=0}^k \binom{k}{j} \binom{k+1}{j+1} + \sum_{j=1}^{k+1} \binom{k}{j-1} \binom{k+1}{j}$$

$$= 2 \cdot 4^k - 2 \sum_{j=0}^k \binom{k}{j} \binom{k+1}{j+1} + \sum_{j=0}^k \binom{k}{j} \binom{k+1}{j+1}$$

$$= \boxed{2 \cdot 4^k - \sum_{j=0}^k \binom{k}{j} \binom{k+1}{j+1}}$$

$$S_{k+1} = 2 \cdot 4^k + \sum_{j=0}^{k+1} \binom{k}{j} \binom{k+1}{j} - \sum_{j=0}^k \binom{k}{j} \binom{k+1}{j+1}$$

$\binom{k}{k+1} = 0$

$$= 4 \cdot 4^k + \sum_{j=0}^k \binom{k}{j} \binom{k+1}{j} - \sum_{j=0}^k \binom{k}{j} \binom{k+1}{j+1}$$

$$\sum_{j=0}^k \binom{k}{j} \left[\binom{k+1}{j} - \binom{k+1}{j+1} \right]$$

$$\text{Claim} = 0$$

$$\sum_{j=0}^k \binom{k}{j} \binom{k+1}{j} - \sum_{j=0}^k \binom{k}{j} \binom{k+1}{j+1}$$

$$\sum_{j=0}^k \binom{k}{\underline{k-j}} \binom{k+1}{\underline{k+1-j}}$$

Set $j' = k-j$

$$\sum_{j'=0}^k \binom{k}{j'} \binom{k+1}{j'+1}$$

cancel each other

$$\therefore S_{k+1} = 4 \cdot 4^k = 4^{k+1} \quad (\text{done by induction})$$

$$\therefore \text{Prob.} = \frac{4^n}{2^{2n+1}} = \frac{1}{2}$$

$\uparrow$
 from Ω

3rd Approach



Let $E \subseteq \Omega$ be the "favourable" event.

$$\omega \in E^c \Leftrightarrow \# \text{heads in Mary} \leq \# \text{heads in John for } \omega$$

$(h_M) \qquad \qquad \qquad (h_J)$

$$\Leftrightarrow -h_J \leq -h_M \text{ for } \omega$$

$$\Leftrightarrow n+1-h_J \leq n+1-h_M \text{ for } \omega$$

$$\Leftrightarrow \underbrace{n-h_J}_{\# \text{Tails in John}} < \underbrace{n+1-h_M}_{\# \text{Tails in M}} \text{ for } \omega$$

Let $f: \Omega \rightarrow \Omega$

$$f(x_1, x_2, \dots, x_{2n+1}) = (\phi(x_1), \phi(x_2), \dots, \phi(x_{2n+1}))$$

where $\phi(x) = \begin{cases} H & \text{if } x = T \\ T & \text{if } x = H \end{cases}$ } "Flipping" operator

From eqv. statement just now

⋮

$$\Leftrightarrow \# \text{ Tails in John} < \# \text{ Tails in Mary for } w$$

$$\Leftrightarrow \# \text{ Heads in Mary} > \# \text{ Heads in John for } f(w)$$

$$\Leftrightarrow f(w) \in E$$

$$\therefore w \in E^c \iff f(w) \in E^c$$

~~$$(w \in f^{-1}(E))$$~~

$$\therefore |E| = |E^c|$$

Note: When $f: \Omega \rightarrow \Omega$ is ~~not~~ inj, ~~for~~

$(x \in X \Rightarrow f(x) \in Y)$ would imply $|X| \leq |Y|$
 $X, Y \subseteq \Omega$

$$\therefore |\Omega| = |E| + |E^c| \quad (\text{Note that } E, E^c \text{ partition } \Omega)$$

$$\Rightarrow |E| = \frac{|\Omega|}{2}$$

$$\text{Prob.} = \frac{|E|}{|\Omega|} = \frac{|\Omega|/2}{|\Omega|} = \frac{1}{2}.$$